

POWER ELECTRONICS

Introduction

What is Power Electronics

- Study, design and applications of electronic circuits and devices to *effectively control and convert* electrical power.
- Uses: Power Semiconductor Devices (e.g. diodes, transistors, thyristors)
- Key Functions: AC to DC (rectification), DC to AC (inversion), DC to DC (choppers), and AC to AC (cycloconverters), regulation (voltage, current and/or power), motor drives, power supplies, switching and amplification.
- Applications: Motor drives, robotics, renewable energy, power supplies, electric vehicles and transportation, power systems.
- Components: switches (diodes, transistors, thyristors), passive elements (capacitors, inductors, resistors), control circuits (DSP, PWM)

About the Course

- Techniques required in the
 - design,
 - analysis, and
 - fabrication of power electronic circuits and systems.

About the Course

- Operational Amplifiers characteristics, universal analysis technique.
- Composite amplifier, suspended supply amplifier,
- Power calculations
- Linear power amplifier, audio amplifier
- Class A and Class B amplifier, amplifier protection
- Power switches, power supplies
- Buck and boost regulators, flyback converters and rectifiers
- Thyristors, proportional power circuits
- Three-phase converters

OPERATIONAL AMPLIFIERS

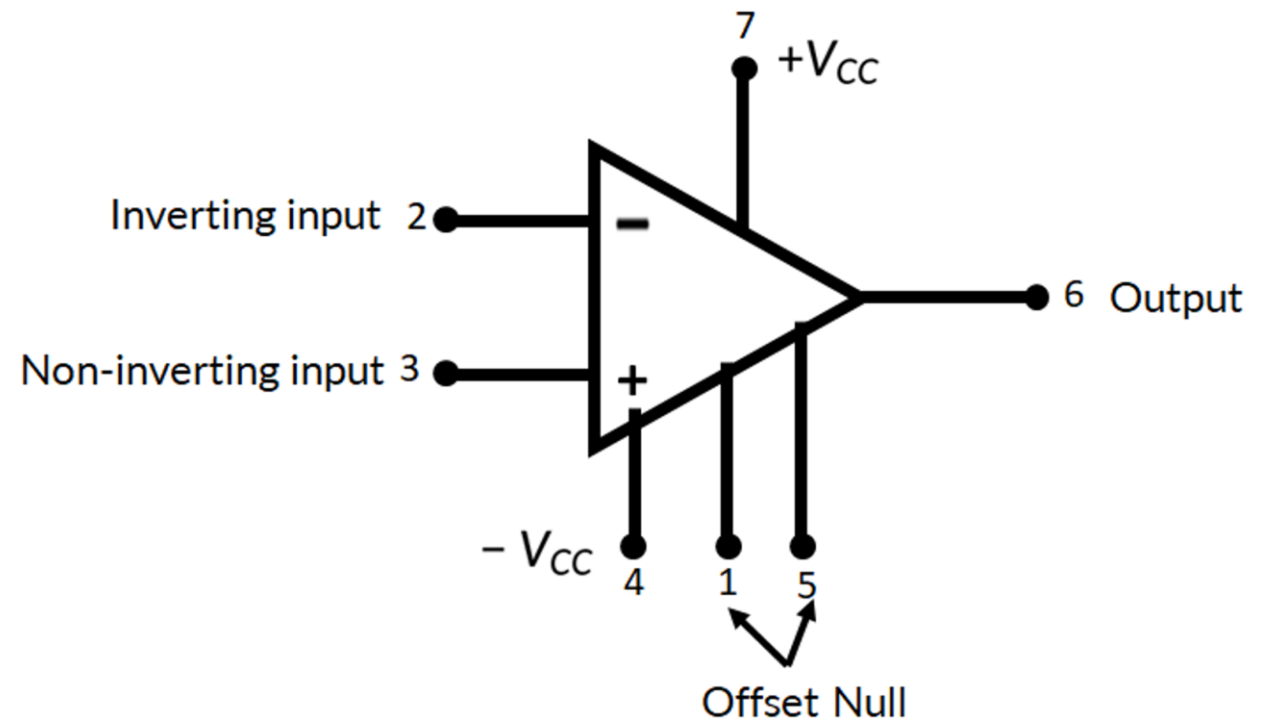
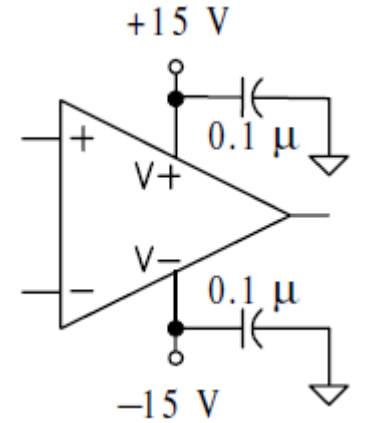
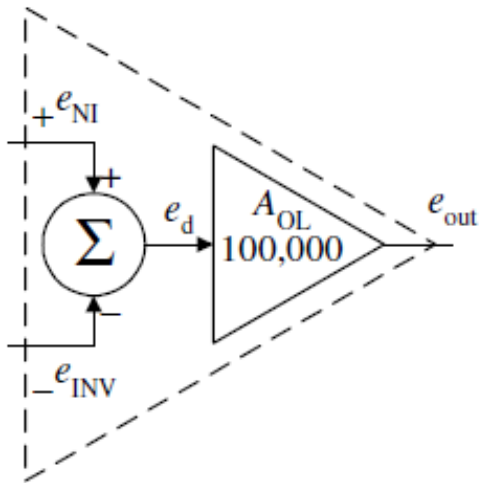
Characteristics and Analysis

Operational Amplifier

- High-gain, electronic voltage amplifier
- Used in analog electronics
- Amplifies the voltage difference between its two input terminals
- Functions: signal conditioning, filtering, mathematical operations, analog to digital conversion.

Operational Amplifier

- 741 IC
- Supply voltage set the limits
- Coupling capacitors
- High input impedance $\geq 1 \text{ M}\Omega$
- High open loop gain $A_{OL} \gg 1000$
- Equations

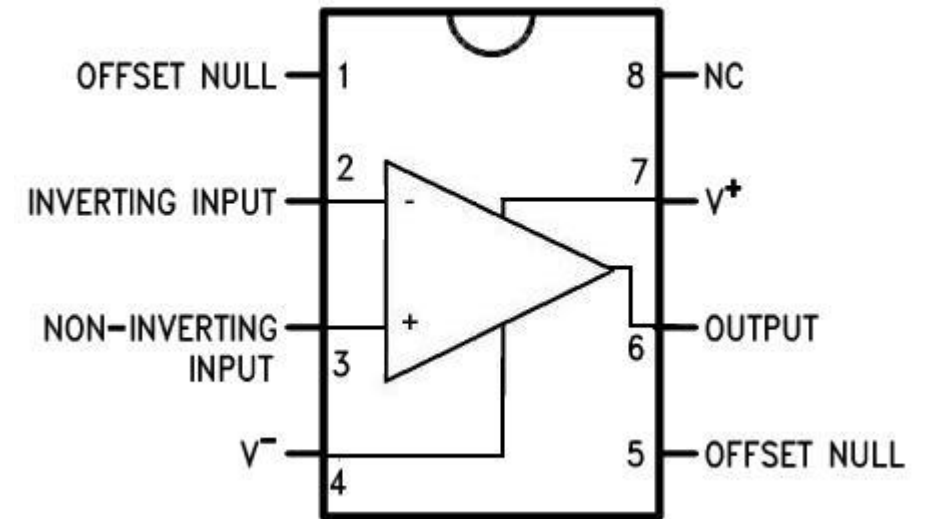


How Actually It Looks Like?

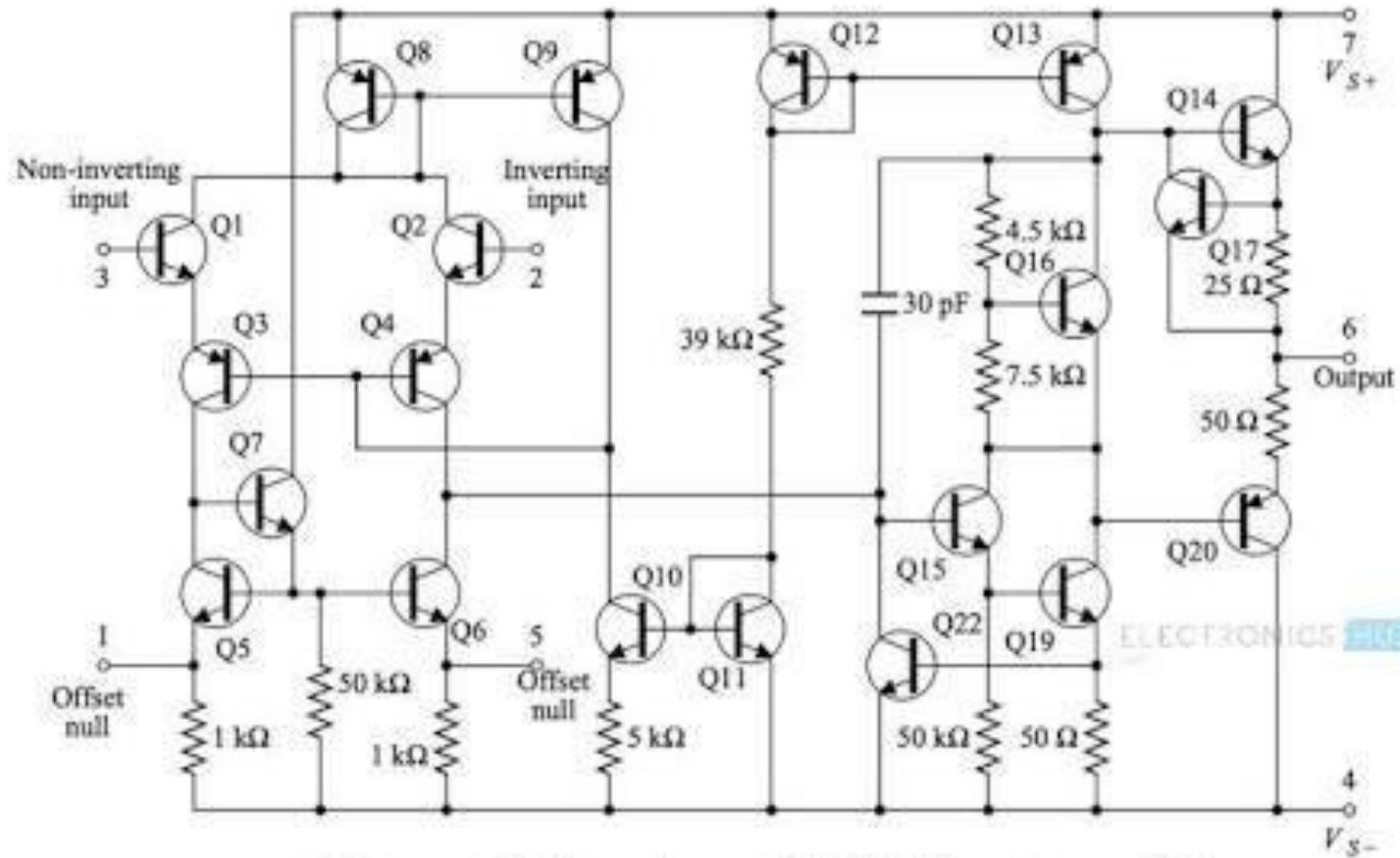
- LM 741



LM741 Pinout Diagram

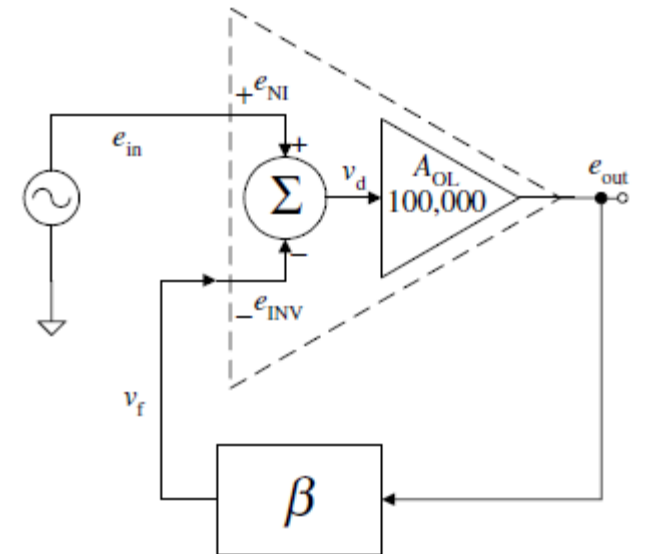


Internal Circuitry of 741 Op-Amp IC



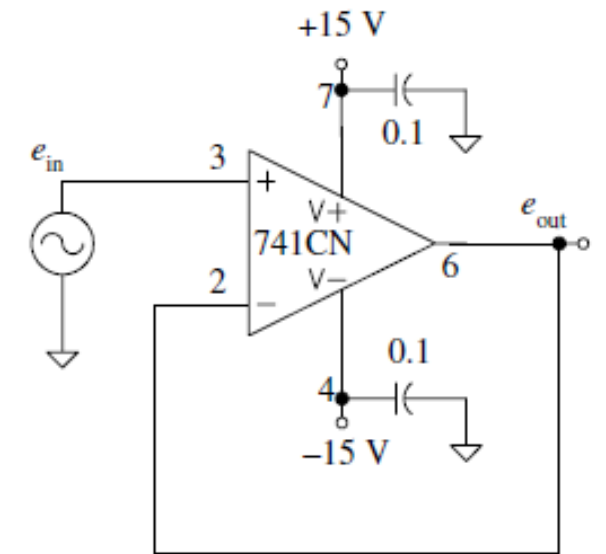
Negative Feedback

- Limitation of open loop op-amp?
- Output *must be fed back* to the inverting input.
- Feedback ratio is β , where $0 \leq \beta \leq 1$
- Closed Loop Gain $A_{CL} \approx \frac{1}{\beta}$



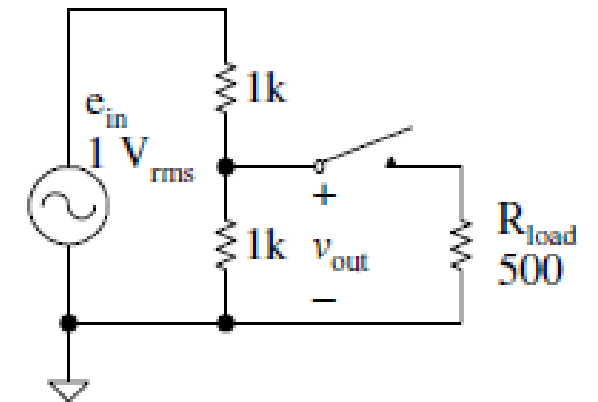
Unity Feedback / Voltage Follower

- $\beta = 1$
- Op-amp internal impedance?
- Load can draw as much current (within in op-amp current limit) as required without loading down the output voltage
- Voltage follower as *buffer*



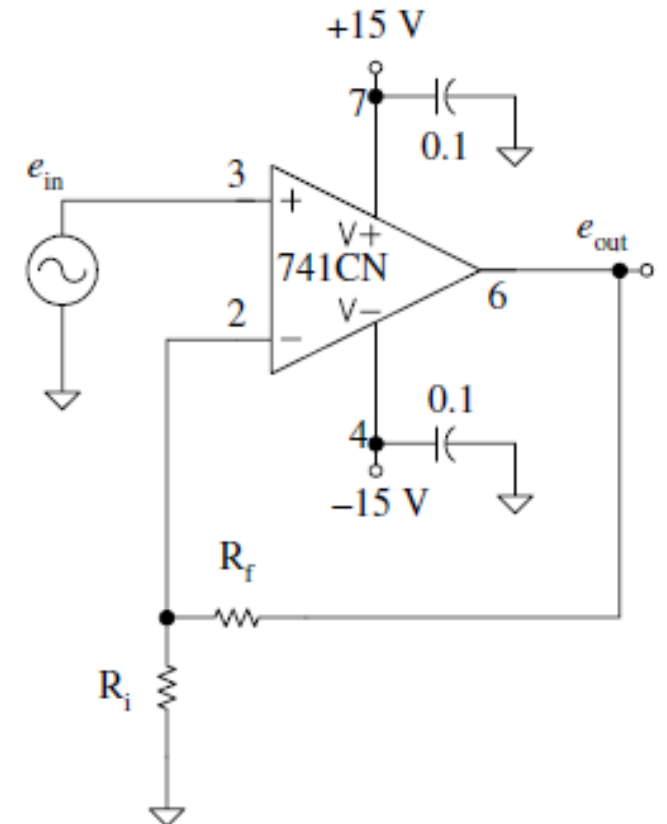
Example 1:

- Calculate v_{out} with the switch open (no load), and with the switch closed (500 Ω load).
- Repeat the calculation using a voltage follower as buffer between voltage divider and the load.



Noninverting Amplifier

- To increase the gain, reduce the negative feedback using voltage divider
- $A_{CL} = 1 + \frac{R_f}{R_i}$
- The gain is positive
- The input and output voltages are in phase (non-inverting)
- Smallest gain is 1 that turns it into voltage follower

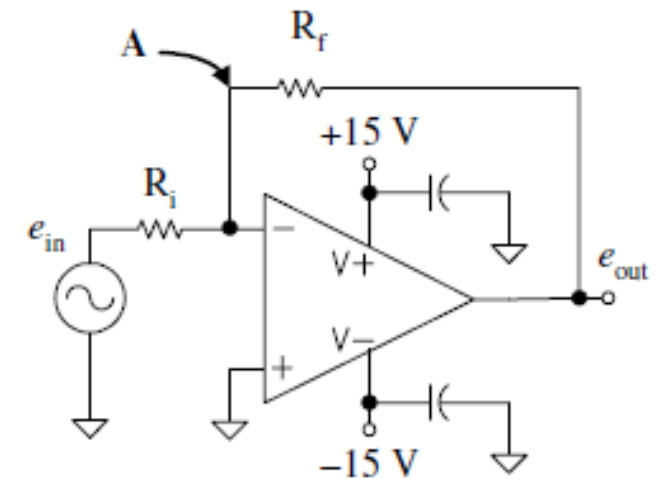
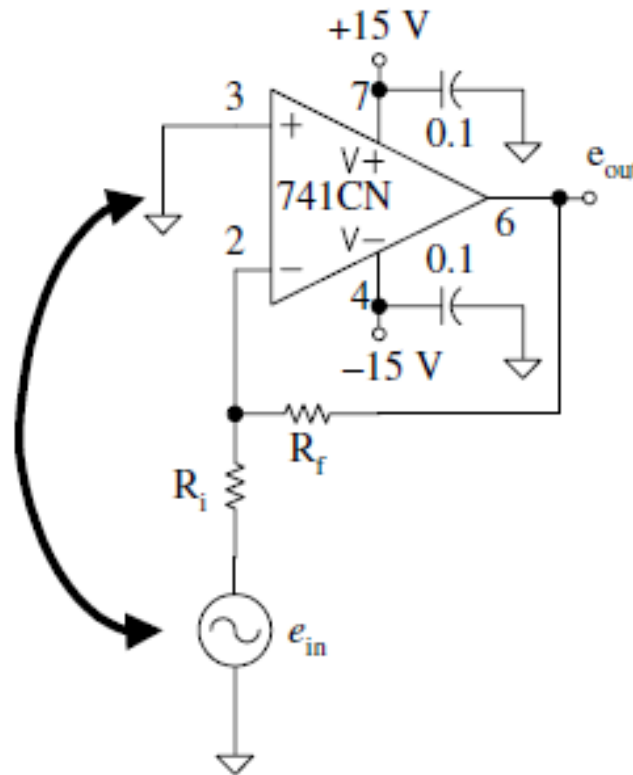


Example 2

- Design a noninverting amplifier with a gain of 100. Pick $R_f = 100\text{ k}\Omega$

Inverting Amplifier

- Reversing the input connections reverses the phase relationship between input and output voltages
- Example

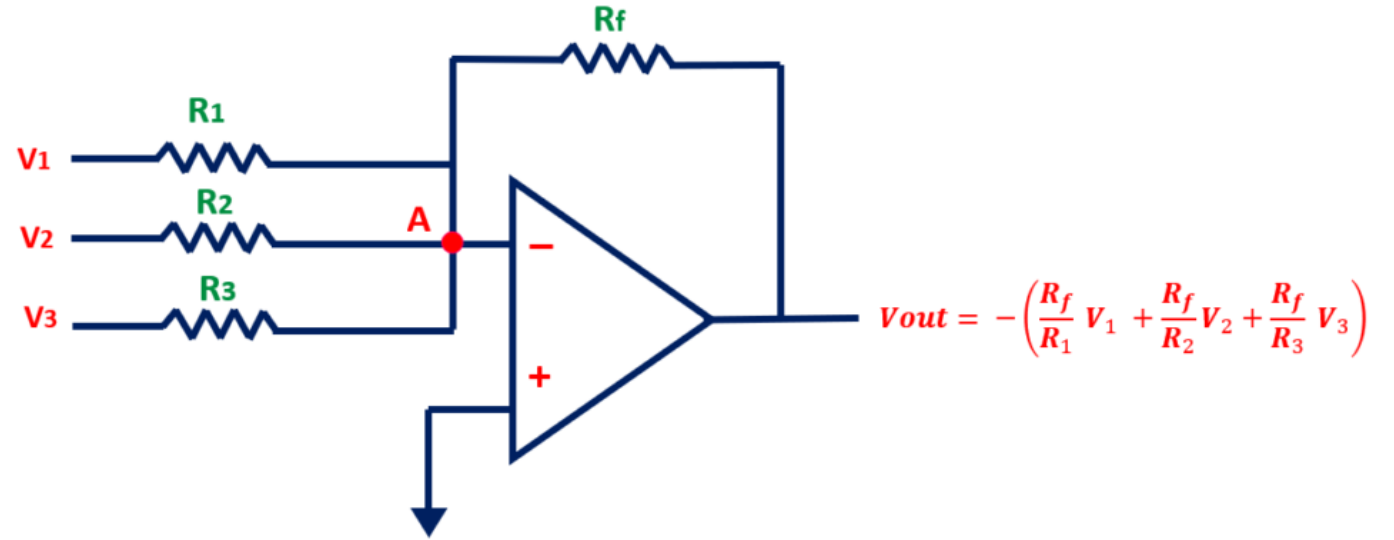
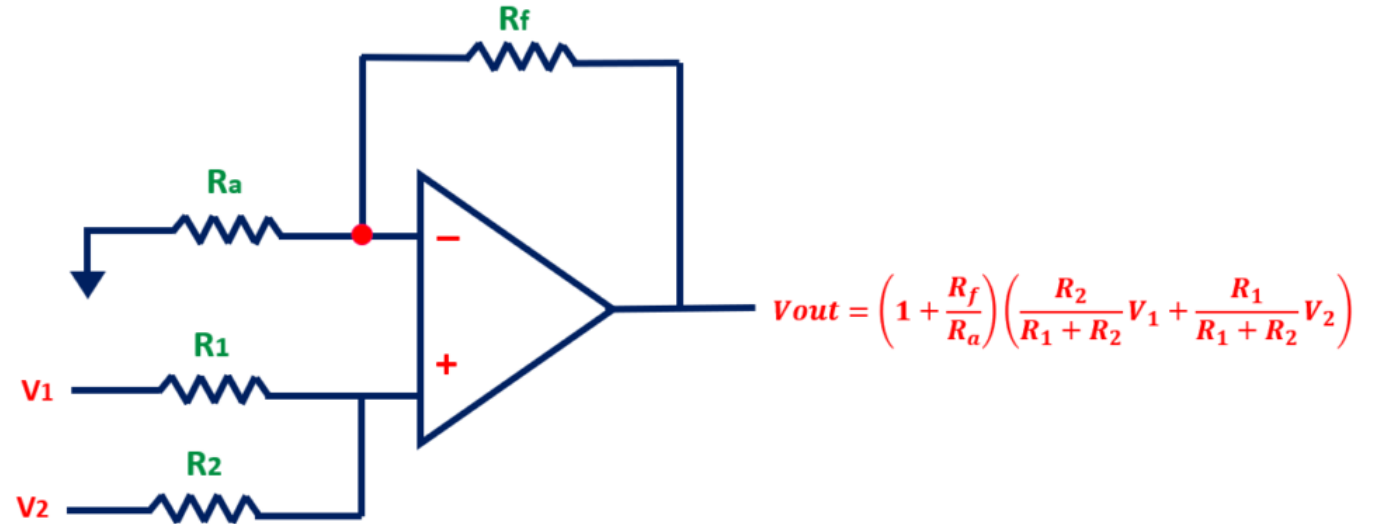


Universal Analysis Technique

- Find voltage on the noninverting input pin
- Find voltage on the inverting input pin
- Calculate current through each of the input resistor
- Calculate current through the feedback resistor using KCL
- Calculate voltage drop across feedback resistor
- Calculate output voltage using KVL

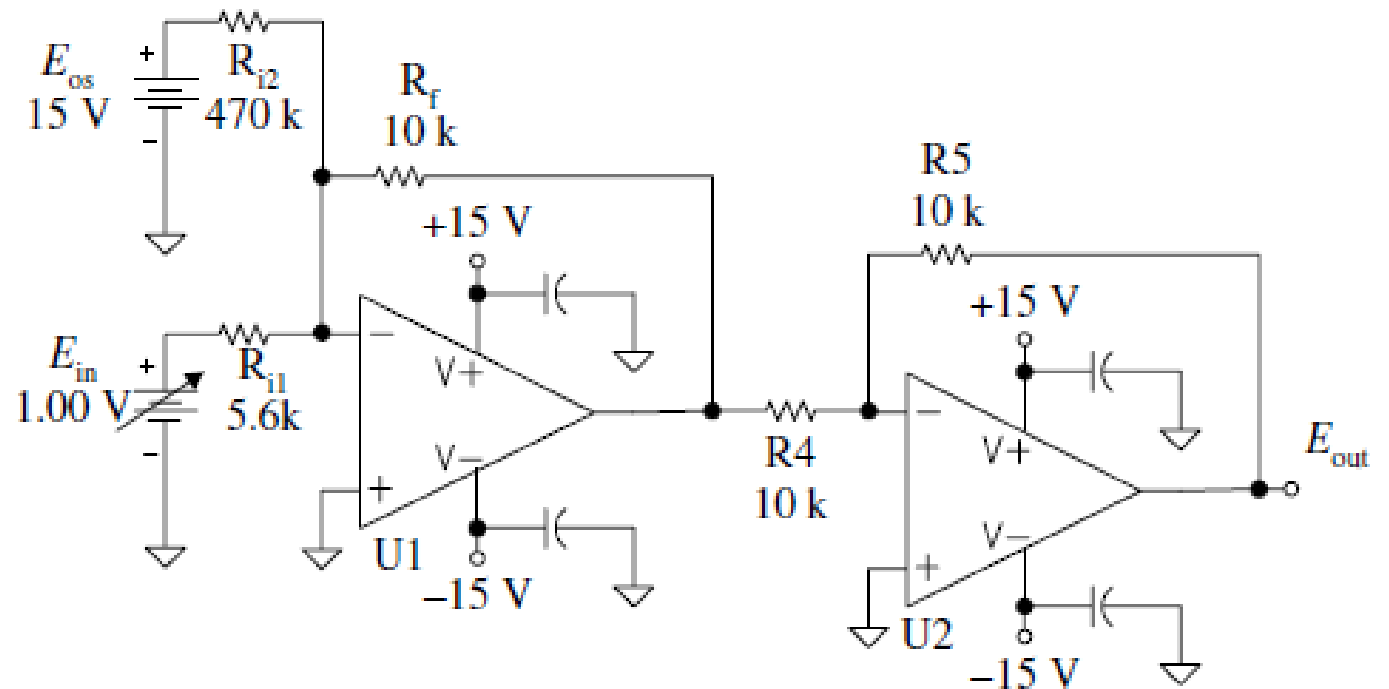
Summer Amplifier

- Used to add analog signals
- Inverting summer
- Noninverting summer



Example 3

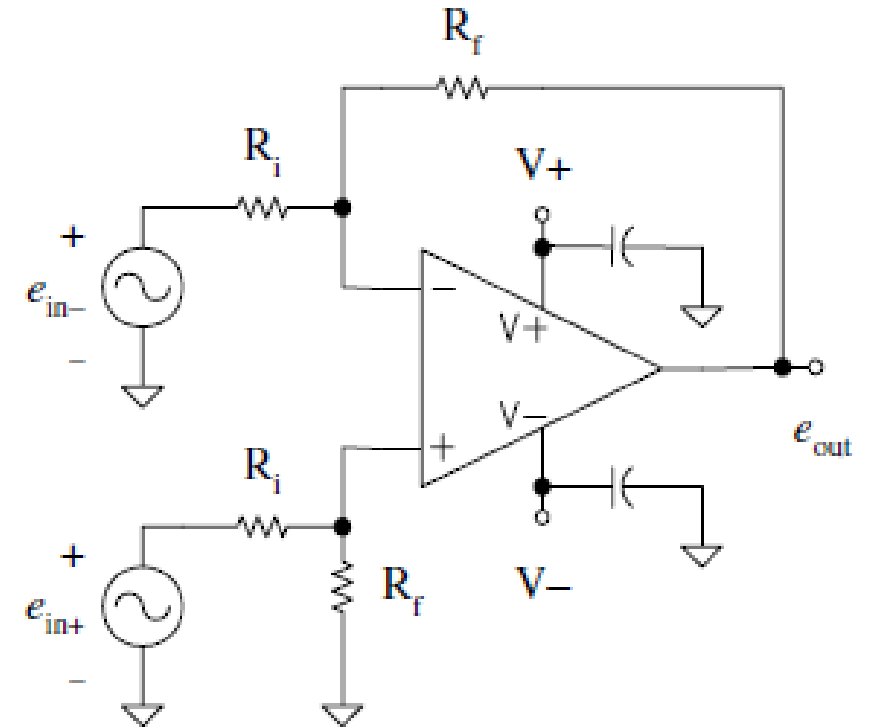
1. Calculate the voltage at the output of the circuit.
2. Calculate the voltage at the output if $E_{in} = 0$ V
3. Calculate new values for R_{i1} and R_{i2} so that $E_{in} = -0.1$ V, $E_{out} = 0$ V
4. Calculate new values for R_{i1} and R_{i2} so that $E_{in} = 0.4$ V, $E_{out} = 5$ V



Difference Amplifier

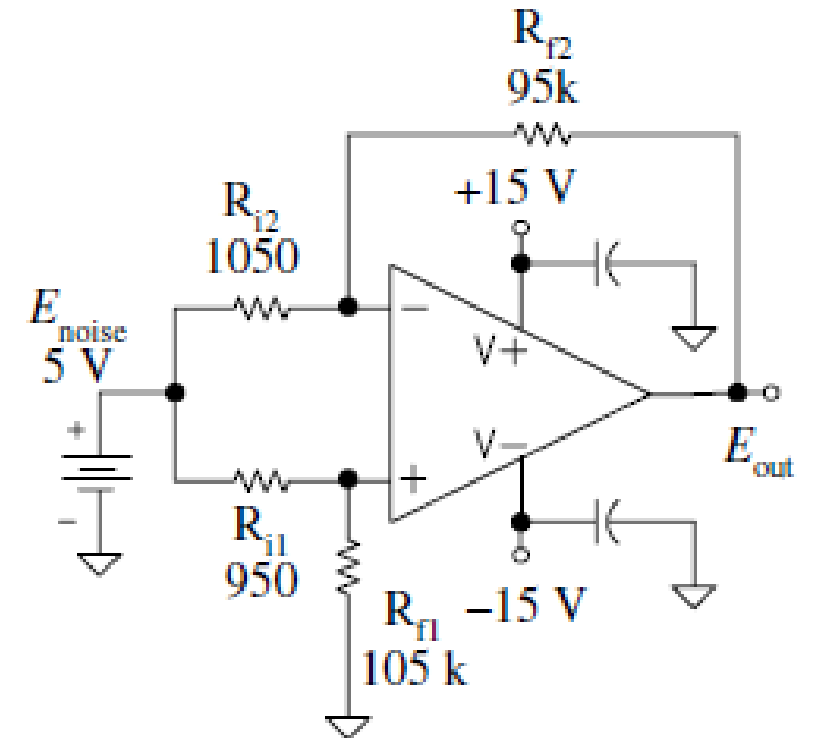
- It has two inputs.
- Output is the difference between two inputs multiplied by the fixed gain.

- $$e_{out} = \frac{R_f}{R_i} (e_{in+} - e_{in-})$$



Example 4

1. Calculate the output voltage for the circuit shown in figure below
2. Repeat the calculations with $R_{i1} = 1050 \Omega$, $R_{i2} = 950 \Omega$, $R_{f1} = 95 \text{ k}\Omega$, and $R_{f2} = 105 \text{ k}\Omega$

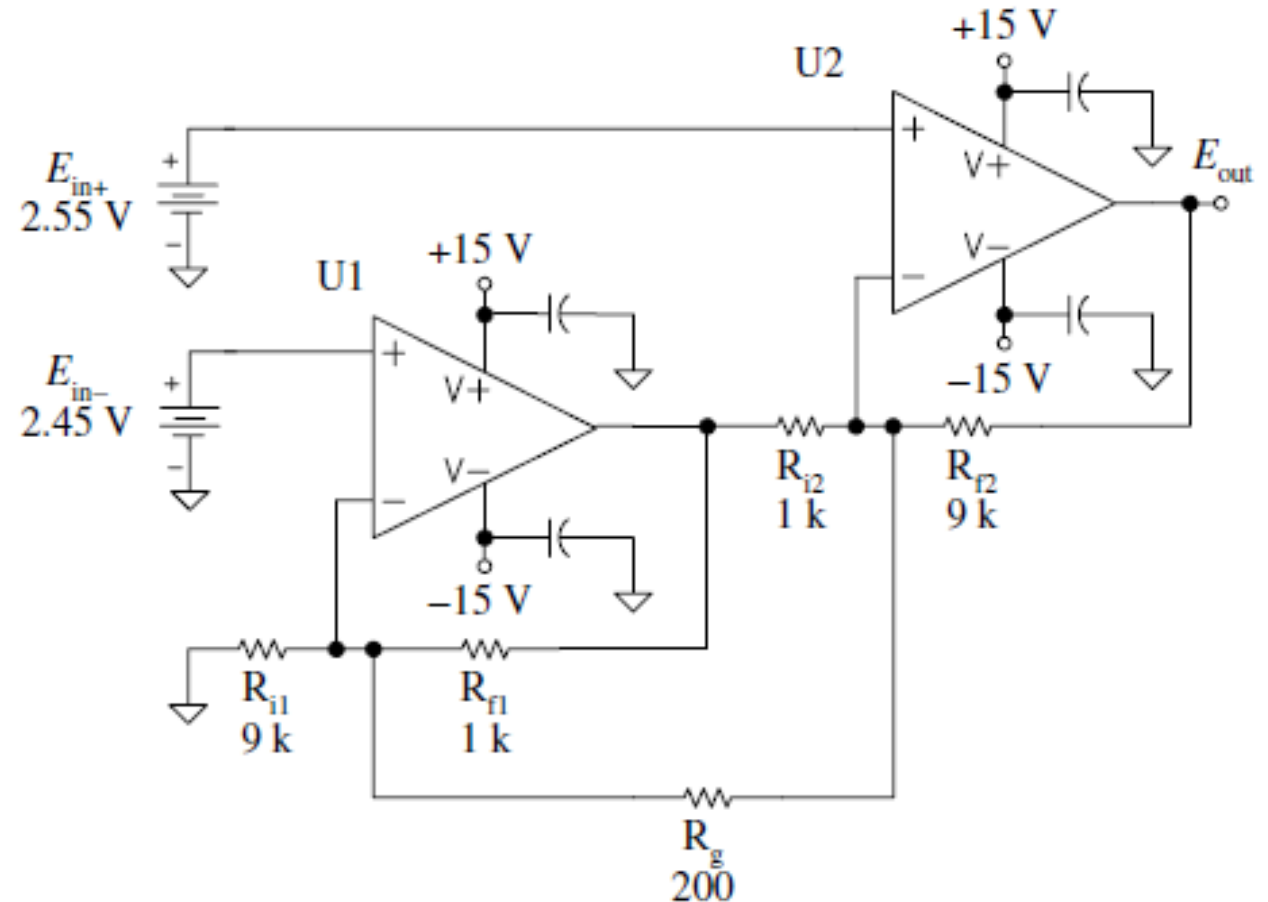


Instrumentation Amplifier

- Mismatch issue with the difference amplifier.
- Two op-amps instrumentation amplifier.
- No mismatch problem in two op-amps instrumentation amplifier
- Gain is adjusted by a single resistor R_g
- Each input is directly connected to the noninverting input pin.

Example 5

1. Analyze the circuit for all currents and voltages.
2. Repeat the problem with $R_g = 400 \Omega$



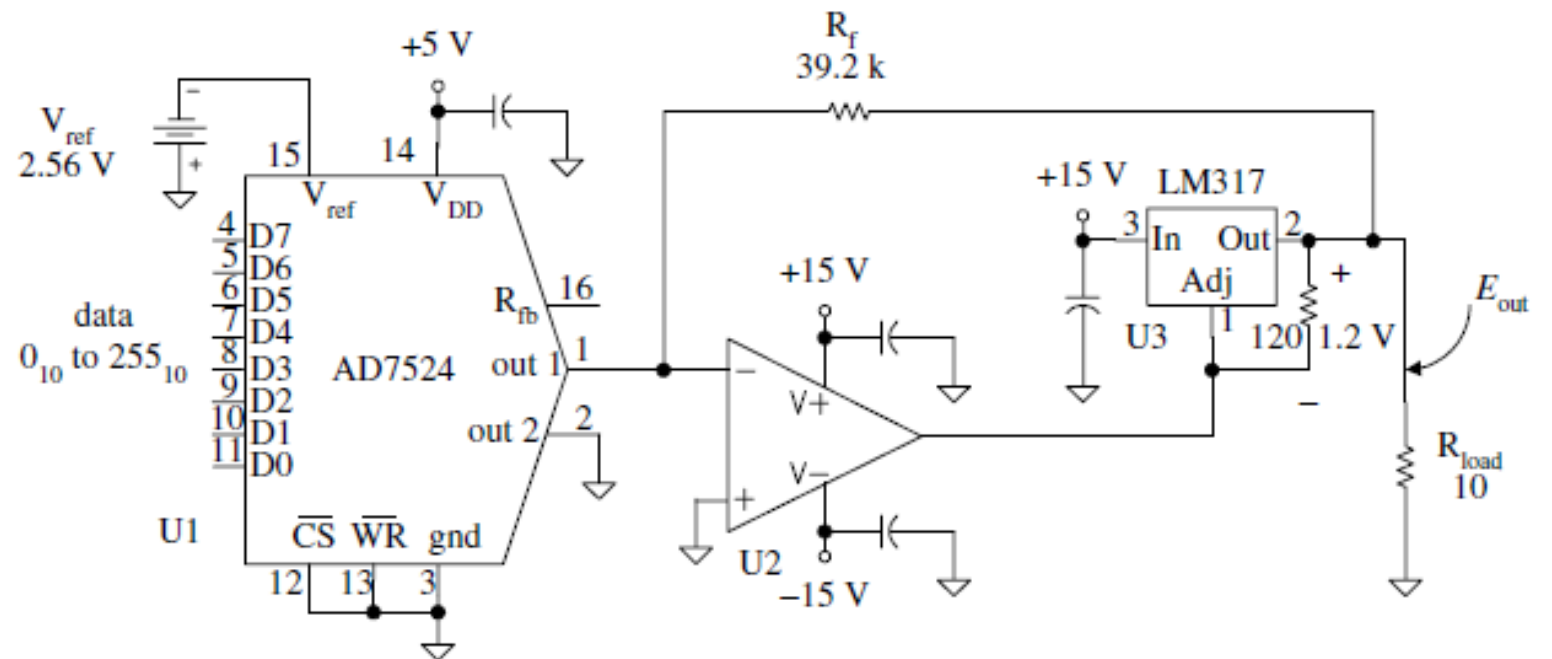
Composite Amplifier

- Composes several amplifiers
- Overcomes the weaknesses and exploits the strengths
- Amplifier within a negative feedback of the other

Example 6

1. Calculate all the currents and voltages in the composite amplifier when the data is 128_{10} . Also determine the power delivered to the load and the power dissipated by LM317.
2. Calculate the load voltage and op-amp's output voltage with data = 15_{10}

AD7524 output current: $I_{out} = \frac{V_{ref}}{10 \text{ k}\Omega} \times \frac{code}{256_{10}}$

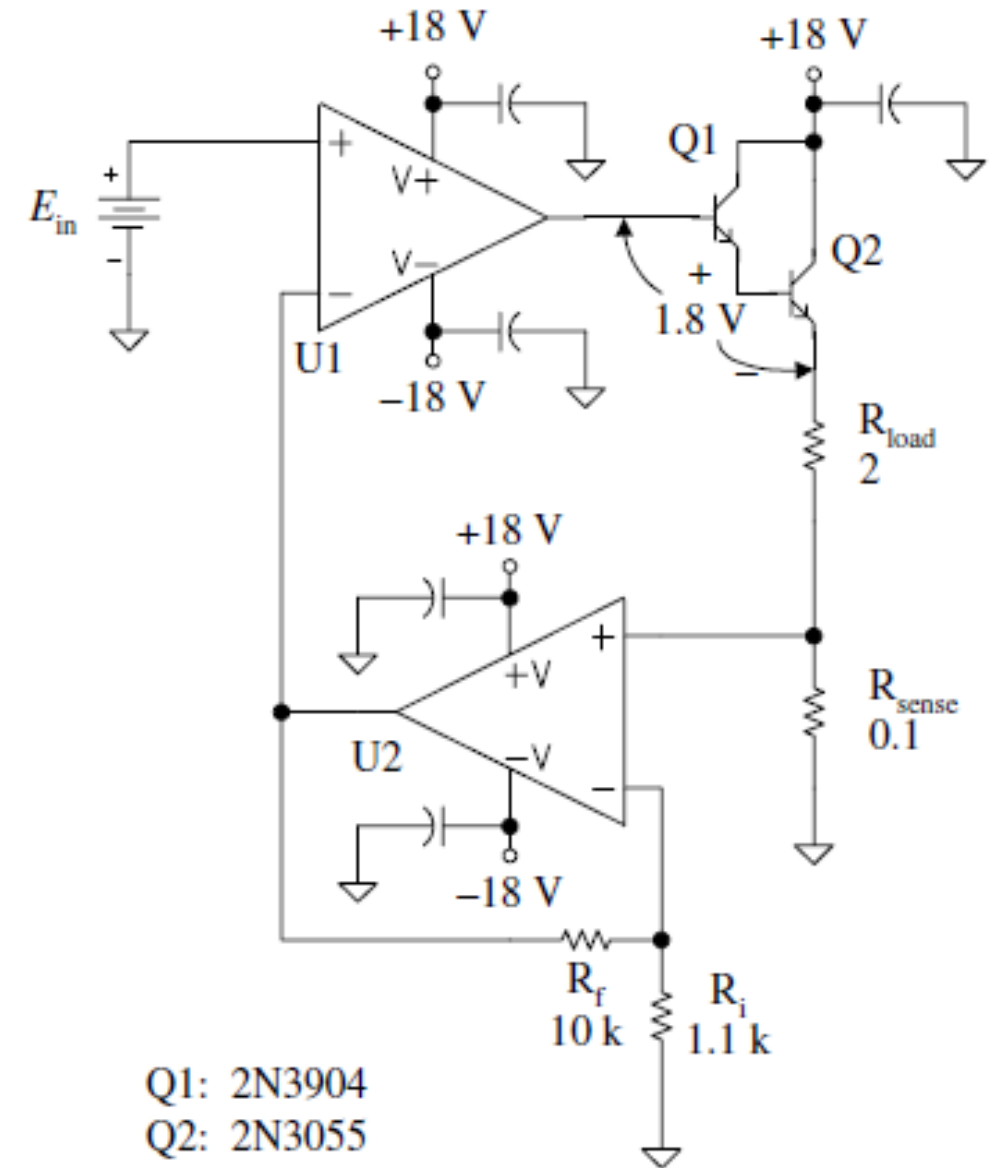


Active Feedback Composite Amplifier

- Amplifier as part of negative feedback loop
- Allows amperes of currents to a low resistance load with high precision of op-amp
- Low priced

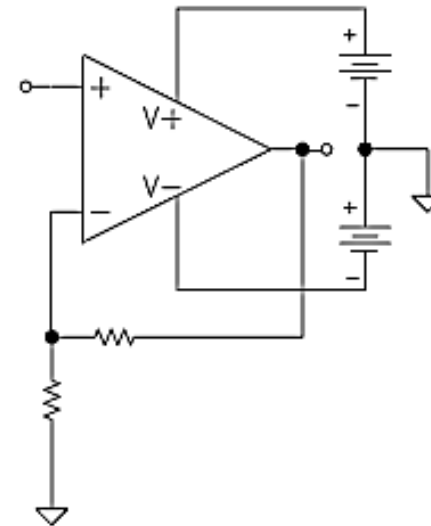
Example 7

Calculate all the currents and voltages in the composite amplifier with an active feedback where $E_{in} = 4$ V.

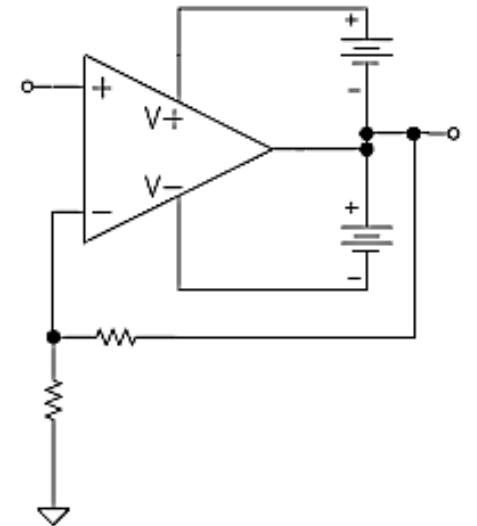


Suspended Supply Amplifier

- Supplies are suspended instead of fixed (dynamically adjusted based on amplifier's output or load requirements)
- Minimal headroom voltage
- Reduced unnecessary power dissipation and hence reduced heat loss
- Allows amplifier to efficiently handle signals over a wide range of power levels



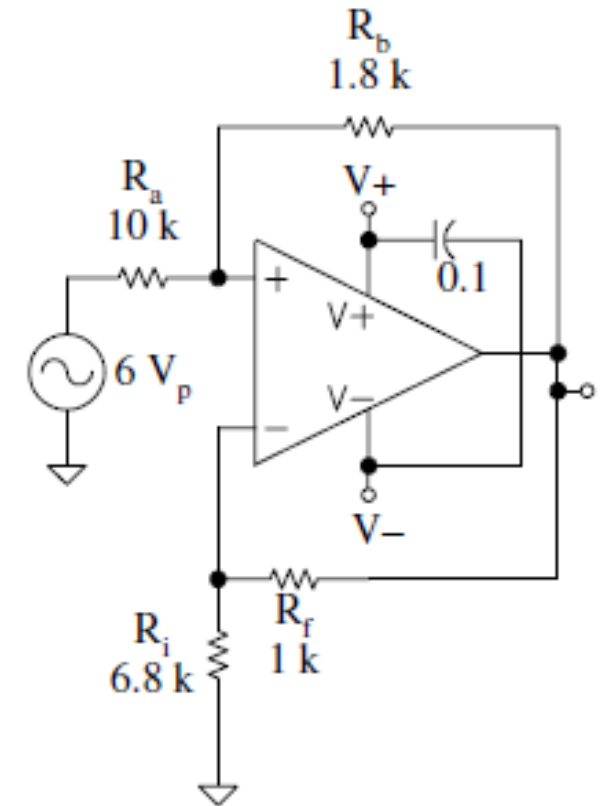
(a) Traditional grounded supplies



(b) Suspended supplies

Example 8

Apply the six steps of the universal analysis technique to calculate the output from the amplifier shown below. Consider the $V_{margin} = 8\text{ V}$.



Reference

Power Electronics, Principles and Applications
J. Michael Jacob

Power Electronics
Ned Mohan